Using the Law of Conservation of Energy to solve problems

Worked solutions + full notes

Use g = 9.8 m s^{-2} throughout unless stated otherwise.

Core principle

Energy cannot be created or destroyed. It can be transferred between energy stores/types and between a system and its surroundings. When we choose a suitable system, the total energy before an event equals the total energy after the event.

Total energy before = Total energy after

For this lesson the important AP Physics 1 quantities are K, ΔU_g, U_g, U_s, and ΔU_{int}. We deliberately do not use the work-energy theorem here. AP Physics 1 focuses on mechanical energy, while students should also recognise dissipation to thermal/internal energy or sound.

Task 1 · Main energy transfers

Small dense ball falling: Gravitational → Kinetic. The ball-Earth system loses gravitational potential energy while the ball gains kinetic energy. Air resistance is assumed small.

Car coasting uphill, engine off: Kinetic → Gravitational. The car slows while the car-Earth system gains gravitational potential energy. Small dissipative losses are neglected for the main-transfer model.

School lift descending at steady speed: Gravitational → Internal. The lift speed is constant, so K is unchanged. The decrease in gravitational potential energy becomes mainly internal energy in the lift/motor/braking system.

Compressed spring launches cart: Elastic → Kinetic. Spring potential energy decreases while cart kinetic energy increases.

Bicycle brakes to rest: Kinetic → Internal. Kinetic energy becomes mainly internal energy in brakes, tyres and surroundings.

Bouncy ball compresses on impact: Kinetic → Elastic. During compression, kinetic energy is stored temporarily as elastic potential energy; some is also transferred to internal energy and sound.

Task 2 · Simple conservation

A. Lightbulb

• Energy supplied each second = 100 J.

• Internal-energy increase each second = 75 J.

• Remaining energy = 100 − 75 = 25 J.

Answer: 25 J each second is transferred mainly as radiation (light).

B. Falling ball

• Gravitational potential energy decreases by 250 J.

• Kinetic energy increases by 200 J.

• Unaccounted energy = 250 − 200 = 50 J.

Answer: 50 J has been transferred mainly to internal energy, with possible sound as well.

C. Elastic band

• At the highest point, speed is zero, so K = 0.

• Losses are neglected.

• Gain in U_g = 25 J must equal the decrease in U_s.

Answer: initial elastic potential energy = 25 J.

Task 3 · 28 g ball falling 78 cm

Given: m = 28 g = 0.028 kg; initial speed $v_0 = 0$; vertical drop = 78 cm = 0.78 m.

(a) Change in vertical position

Take upward as positive:

$$\Delta y = y_f - y_i = -0.78 \text{ m}$$

(b) Change in gravitational potential energy

$\Delta U_g = mg\Delta y = (0.028)(9.8)(-0.78) = -0.214 \text{ J} \approx -0.21 \text{ J}$.

(c) Meaning of the sign

The negative sign means the gravitational potential energy of the ball-Earth system has decreased.

(d) Speed immediately before impact

Using $v^2 = v_0^2 + 2a\Delta y$ with downward quantities taken consistently:

$v^2 = 2(9.8)(0.78) = 15.288$, so $v = 3.91 \text{ m s}^{-1} \approx 3.9 \text{ m s}^{-1}$.

(e) Kinetic energy

$K = \frac{1}{2}mv^2 = \frac{1}{2}(0.028)(3.91)^2 = 0.214 \text{ J} \approx 0.21 \text{ J}$.

(f) Compare the changes

The ball gains 0.21 J of kinetic energy while the ball-Earth system loses 0.21 J of gravitational potential energy. With negligible drag, $\Delta K = -\Delta U_g$.

Task 4 · Before-and-after accounting

A useful general bookkeeping statement is:

$$K_i + U_{g,i} + U_{s,i} + U_{int,i} = K_f + U_{g,f} + U_{s,f} + U_{int,f}$$

For a ball dropped from rest with negligible drag, choose the final height as the zero of U_g. Then $K_i = 0$, $U_{g,f} = 0$, and $\Delta U_{int} \approx 0$, leaving $U_{g,i} = K_f$.

Task 5 · Bouncy ball practical

At release, the ball is at rest, so its mechanical energy relative to the floor is approximately:

$$E_{before} \approx mgh_1$$

At the top of the first rebound the ball is again momentarily at rest:

$$E_{rebound} \approx mgh_2$$

The mechanical energy not recovered is therefore:

$$\Delta E_{mech,not\ recovered} \approx mg(h_1 - h_2)$$

This energy is mainly transferred to internal energy of the ball and floor, plus sound and small air-resistance losses.

Percentage mechanically recovered ≈ 100 × h_2/h_1. Notice that mass cancels from this ratio.

Optional timing method: if Δt is the time between the first and second impacts, the upward flight lasts approximately Δt/2. Using $h = \frac{1}{2}gt^2$ for the upward/downward symmetry gives $h_2 \approx g(\Delta t)^2/8$. Manual stopwatch reaction time makes this much less precise than video timing, so repeated trials are essential.

Task 6A · 1.2 kg block on frictionless incline

Geometry: vertical drop = 0.30 m, distance along plane = 0.50 m.

(a) $\sin\theta = 0.30/0.50 = 0.60 \rightarrow \theta = 36.9°$.

(b) $F_{\parallel} = mg\sin\theta = (1.2)(9.8)(0.60) = 7.06\ N \approx 7.1\ N$.

(c) $a = F_{net}/m = 7.056/1.2 = 5.88\ m\ s^{-2}$.

(d) $v^2 = v_0^2 + 2as = 0 + 2(5.88)(0.50) = 5.88$, so $v = 2.42\ m\ s^{-1}$.

(e) $\Delta U_g = mg\Delta y = (1.2)(9.8)(-0.30) = -3.53\ J$.

(f) $K = \frac{1}{2}mv^2 = \frac{1}{2}(1.2)(2.4249)^2 = 3.53\ J$. The energy and dynamics routes agree.

Task 6B · Projectile from 12 m building

Given: m = 0.25 kg; v_0 = 15 m s^{-1}; launch angle = 30°.

(a) $v_{x,0} = v_0\cos30° = 15(0.8660) = 13.0$ m s^{-1}.

(b) $v_{y,0} = v_0\sin30° = 15(0.5) = 7.5$ m s^{-1}.

(c) At maximum height $v_y = 0$. Using $v_y^2 = v_{y,0}^2 + 2a_y\Delta y$:

$0 = 7.5^2 - 2(9.8)\Delta y \rightarrow \Delta y = 2.87$ m.

(d) $K_i = \frac{1}{2}mv_0^2 = \frac{1}{2}(0.25)(15^2) = 28.1$ J.

(e) With no air resistance, v_x remains 13.0 m s^{-1}.

(f) At the top, $v_y = 0$, so speed = $|v_x|$ = 13.0 m s^{-1}.

(g) $K_{top} = \frac{1}{2}(0.25)(12.990^2) = 21.1$ J.

(h) $\Delta K = K_{top} - K_i = 21.094 - 28.125 = -7.03$ J.

(i) $\Delta U_g = mg\Delta y = (0.25)(9.8)(2.8699) = +7.03$ J.

As expected, $\Delta K + \Delta U_g = 0$.

(j) From the roof to the ground, $\Delta y = -12$ m, so $\Delta U_g = (0.25)(9.8)(-12) = -29.4$ J.

(k) Therefore the kinetic-energy change from launch to ground is +29.4 J.

(l) $K_{ground} = 28.125 + 29.4 = 57.525$ J. Then $v = \sqrt{(2K/m)} = \sqrt{[2(57.525)/0.25]} = 21.5$ m s^{-1}.

Task 7 · General gravitational potential energy

For large distances use:

$$U_g = -Gm_1m_2/r$$

with zero potential energy at infinite separation.

Data: $G = 6.67 \times 10^{-11}$ N m² kg^{-2}; $M_E = 5.98 \times 10^{24}$ kg; $m = 65$ kg; $R_E = 6380$ km $= 6.38 \times 10^{6}$ m.

(a)(i) At Earth's surface:

$U_g = -[(6.67\times10^{-11})(5.98\times10^{24})(65)]/(6.38\times10^{6}) = -4.06 \times 10^{9}$ J $\approx -4.1 \times 10^{9}$ J.

(a)(ii) At $2R_E$, the denominator doubles, so $U_g = -2.03 \times 10^{9}$ J $\approx -2.0 \times 10^{9}$ J.

(a)(iii) Increase $= U_{g,2} - U_{g,1} = (-2.03\times10^{9}) - (-4.06\times10^{9}) = +2.03 \times 10^{9}$ J.

(a)(iv) At infinity $U_g = 0$, so escape requires $0 - (-4.06\times10^{9}) = 4.06 \times 10^{9}$ J.

(b) Constant-g estimate for a rise of 6380 km:

$\Delta U_g = mg\Delta y = (65)(9.8)(6.38\times10^{6}) = 4.06 \times 10^{9}$ J.

This is about twice the correct increase to $r = 2R_E$. The approximation fails because g decreases substantially over an Earth-radius change.

Task 8 · Falling ball with drag

Given: m = 520 g = 0.520 kg; h = 14 m.

(a) Gravitational potential energy decrease magnitude:

mgh = (0.520)(9.8)(14) = 71.344 J ≈ 71 J.

(b) Ideal case: all 71.344 J becomes kinetic energy, so K = 71.344 J.

(c) $K = \frac{1}{2}mv^2 \rightarrow v = \sqrt{(2K/m)} = \sqrt{[2(71.344)/0.520]} = 16.6 \text{ m s}^{-1} \approx 17 \text{ m s}^{-1}$.

(d) Half of the gravitational-energy decrease becomes internal energy. Therefore K = 71.344/2 = 35.672 J.

$v = \sqrt{[2(35.672)/0.520]} = 11.7 \text{ m s}^{-1} \approx 12 \text{ m s}^{-1}$.

(e) If acceleration is constant, $v^2 = 2ah$:

$a = v^2/(2h) = 11.713^2/(28) = 4.90 \text{ m s}^{-2}$ downward.

(f) Weight = mg = (0.520)(9.8) = 5.096 N downward.

Net force = ma = (0.520)(4.90) = 2.548 N downward.

Therefore $F_{drag} = 5.096 - 2.548 = 2.548 \text{ N} \approx 2.5 \text{ N}$ upward.

Task 9 · Generalising by deriving

(a) Starts from rest

Initial gravitational energy lost = final kinetic energy gained:

$$\mathbf{MgH = \tfrac{1}{2}Mv^2}$$

Cancel M and rearrange: $2gH = v^2$, therefore $v = \sqrt{(2gH)}$.

(b) Initial downward speed v_0

Initial K + gravitational energy lost = final K:

$$\mathbf{\tfrac{1}{2}Mv_0^2 + MgH = \tfrac{1}{2}Mv^2}$$

Multiply by 2/M: $v_0^2 + 2gH = v^2$. Therefore $v = \sqrt{(v_0^2 + 2gH)}$.

This is exactly the familiar constant-acceleration equation $v^2 = v_0^2 + 2a\Delta x$ for downward acceleration a = g and downward displacement H.

Task 10 · Projectiles

1. Horizontal launch

$m = 2.5$ kg, $v_0 = 5.0\ \text{m s}^{-1}$, $H = 12$ m.

(a) $K_i = \tfrac{1}{2}(2.5)(5.0^2) = 31.25\ \text{J} \approx 31\ \text{J}$.

(b) Gravitational potential energy decreases by $mgh = (2.5)(9.8)(12) = 294$ J. With negligible air resistance, K increases by 294 J.

(c) $K_f = 31.25 + 294 = 325.25$ J.

$v = \sqrt{(2K/m)} = \sqrt{[2(325.25)/2.5]} = 16.1\ \text{m s}^{-1} \approx 16\ \text{m s}^{-1}$.

2. General horizontal launch

$\tfrac{1}{2}Mv_0^2 + MgH = \tfrac{1}{2}Mv^2 \rightarrow v = \sqrt{(v_0^2 + 2gH)}$.

3. Angled projectile

$m = 0.500$ kg; $v_0 = 12\ \text{m s}^{-1}$; $\theta = 60°$.

(a) $K_i = \tfrac{1}{2}(0.500)(12^2) = 36$ J.

(b) $v_x = 12\cos 60° = 6.0\ \text{m s}^{-1}$.

(c) With no air resistance, v_x stays constant.

(d) At the highest point $v_y = 0$.

(e) Speed at the top is therefore $6.0\ \text{m s}^{-1}$, so $K_{top} = \tfrac{1}{2}(0.500)(6.0^2) = 9.0$ J.

(f) K lost $= 36 - 9 = 27$ J. With no losses, U_g gained $= 27$ J.

(g) $mgh = 27 \rightarrow h = 27/[(0.500)(9.8)] = 5.51\ \text{m} \approx 5.5\ \text{m}$.

Task 11 · Inclined planes

A. Force and motion

(a) $F_{net} = mg \sin 30° = (2.0)(9.8)(0.5) = 9.8$ N down the plane.

(b) $a = F/m = 9.8/2.0 = 4.9$ m s^{-2}.

(c) $v^2 = 2ad = 2(4.9)(0.40) = 3.92 \rightarrow v = 1.98$ m s$^{-1} \approx 2.0$ m s^{-1}.

B. Conservation of energy

(a) $h = d \sin 30° = (0.40)(0.5) = 0.20$ m.

(b) Gravitational potential energy decrease magnitude $= mgh = (2.0)(9.8)(0.20) = 3.92$ J. Therefore $K_f = 3.92$ J.

(c) $\tfrac{1}{2}mv^2 = 3.92 \rightarrow v^2 = 3.92 \rightarrow v = 1.98$ m s^{-1}.

C. Generalise

(a) $h = d \sin\theta$.

(b) Gravitational potential energy decrease magnitude $= Mgd \sin\theta$.

(c) $\tfrac{1}{2}Mv^2 = Mgd \sin\theta$. Cancel M:

$$\mathbf{v = \sqrt{(2gd \sin\theta)}}$$

(d) Mass cancels. This is consistent with the acceleration of a frictionless block on an incline being independent of mass.

Task 12 · Challenge algebra: rough inclined plane

The block starts from rest. The prompt states that acceleration is down the plane, so $\tan\theta > \mu$.

(a) Distance along the plane

$\sin\theta = h/d \rightarrow d = h/\sin\theta$.

(b) Kinetic friction

$N = mg\cos\theta$, so $F_f = \mu mg\cos\theta$.

(c) Net force down the plane

$F_{net} = mg\sin\theta - \mu mg\cos\theta = mg(\sin\theta - \mu\cos\theta)$.

(d) Acceleration

$a = F_{net}/m = g(\sin\theta - \mu\cos\theta)$.

(e) Final speed

$v^2 = 2ad = 2[g(\sin\theta - \mu\cos\theta)](h/\sin\theta)$.

Factor $\sin\theta$ from the bracket:

$v^2 = 2gh[1 - \mu(\cos\theta/\sin\theta)] = 2gh(1 - \mu\cot\theta)$.

Therefore $v = \sqrt{[2gh(1 - \mu\cot\theta)]}$.

(f) Final kinetic energy

$K_f = \frac{1}{2}mv^2 = \frac{1}{2}m[2gh(1 - \mu\cot\theta)] = mgh(1 - \mu\cot\theta)$.

(g) Change in gravitational potential energy

The vertical displacement is $-h$, so $\Delta U_g = -mgh$.

(h) Increase in internal energy using conservation

Choose the final level as $U_{g,f} = 0$. The block starts from rest, so initial mechanical energy = mgh.

$$\mathbf{mgh = K_f + \Delta U_{int}}$$

$\Delta U_{int} = mgh - mgh(1 - \mu\cot\theta) = \mu mgh\cot\theta$.

This final result accounts for the gravitational energy that did not become kinetic energy.

Final checklist

- Identify the dominant energy transfer first.
- State clearly what is initial and final.
- Use total energy before = total energy after.
- Keep signs on ΔU_g consistent with Δy.
- Do not say energy is “lost” without stating where it went.
- Choose force-and-motion or energy according to which gives the cleanest route.